

MohammadReza EskandariNasab

PHD CANDIDATE IN COMPUTER SCIENCE

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🏠 samresume.com | 🔗 linkedin.com/in/samresume | 📄 github.com/samresume

👤 ORCID: 0009-0004-0697-3716

🎓 scholar.google.com/citations?user=oUuVbaUAAAAJ (Articles: 10, Citations: 129)

🔍 researchgate.net/profile/Mohammadreza-Eskandarinasab (Research items: 16, Citations: 128)

Education

Utah State University

Utah, USA

PH.D. IN COMPUTER SCIENCE

Jan 2025 - Expected Aug 2027

- GPA: 4.0/4.0
- Conducting research on generative models, time series analysis, and solar flare prediction.
- Dissertation proposal on denoising diffusion probabilistic models and multimodal learning.
- Supervisor: Dr. Shah Muhammad Hamdi

Utah State University

Utah, USA

M.SC. IN COMPUTER SCIENCE

May 2023 - Dec 2024

- GPA: 4.0/4.0
- Thesis: Supervised Generative Adversarial Networks for Time Series Generation in Embedding Space.
- Supervisor: Dr. Shah Muhammad Hamdi

University of Zanjan

Zanjan, Iran

B.SC. IN COMPUTER ENGINEERING

Aug 2017 - Aug 2021

- GPA: 3.88/4.00 (last two years)
- Project: Car Dealership Web Application using Django and React with a Dynamic Reputation-Weighted Pricing Algorithm.

Research Interests

Pure Machine Learning

- Generative Modeling
- Multivariate Time Series Classification
- Contrastive Learning and Domain Adaptation

Applications of Machine Learning

- Solar Flare Prediction
- Selective Auditory Attention
- Biomedical Signal Processing

Experience

Utah State University

Utah, USA

MACHINE LEARNING RESEARCHER

Sep 2023 - Now

- Developed TIMED, SeriesGAN, and AVATAR, state-of-the-art generative models that improve the quality of time series generation by leveraging denoising diffusion probabilistic models (DDPMs), generative adversarial networks (GANs), and adversarial autoencoders. Accepted as first author to IEEE ICDM 2025, IEEE Big Data 2024, and SIAM SDM 2025. Designed a contrastive representation learning-based classifier and custom data preprocessing pipelines, achieving solar flare classification accuracy that outperforms existing benchmarks. Currently conducting research on multimodal generative learning to develop models for generating and converting space weather time series data.

- Developed FlaPLeT, a full-stack web platform for an NSF-funded project that provides machine learning tools for solar flare classification without requiring coding expertise. The platform offers an intuitive interface, enabling researchers to create accounts, upload datasets, and apply preprocessing, data augmentation, and classification techniques seamlessly. Built with Django, REST APIs, React, PostgreSQL, Celery, Redis, Nginx, Waitress, and machine learning frameworks such as TensorFlow, Keras, and Pandas, FlaPLeT supports end-to-end workflows for solar flare prediction. Users can configure classifiers, fine-tune hyperparameters, and access detailed performance reports. [Architecture Explained (YouTube)] - [Functionality Demo (YouTube)]

Publications

2026

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “FlaPLeT: A full-stack web platform for end-to-end time series data processing and machine learning in solar flare prediction,” *SoftwareX*, vol. 33, p. 102540, 2026. [Online]. Available: <https://doi.org/10.1016/j.softx.2026.102540>, **impact factor: 2.4**

Peer-Reviewed
Journal

2025

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “AVATAR: Adversarial Autoencoders with Autoregressive Refinement for Time Series Generation,” *SIAM International Conference on Data Mining (SDM 25)*, April 2025. [Online]. Available: <https://doi.org/10.1137/1.9781611978520.59>, **acceptance rate: 26.7%**

Peer-Reviewed
Conference

2025

Z. Raeisi, O. Bashiri, **M. EskandariNasab**, *et al.*, “EEG microstate biomarkers for schizophrenia: a novel approach using deep neural networks,” in *Cognitive Neurodynamics*, vol. 19, p. 68, 2025. [Online]. Available: <https://doi.org/10.1007/s11571-025-10251-z>, **impact factor: 3.9**

Peer-Reviewed
Journal

2024

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “Impacts of data preprocessing and sampling techniques on solar flare prediction from multivariate time series data of photospheric magnetic field parameters,” *The Astrophysical Journal Supplement Series*, vol. 275, no. 1, p. 6, Oct. 2024. [Online]. Available: <https://dx.doi.org/10.3847/1538-4365/ad7c4a>, **impact factor: 8.6**

Peer-Reviewed
Journal

2024

M. EskandariNasab, Z. Raeisi, R. A. Lashaki, and H. Najafi, “A GRU-CNN model for auditory attention detection using microstate and recurrence quantification analysis,” *Scientific Reports*, vol. 14, no. 1, p. 8861, Apr. 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-58886-y>, **impact factor: 3.8**

Peer-Reviewed
Journal

2024

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “SeriesGAN: Time Series Generation via Adversarial and Autoregressive Learning,” *IEEE International Conference on Big Data (Big Data)*, Oct. 2024. [Online]. Available: <https://doi.org/10.1109/BigData62323.2024.10825115>, **acceptance rate: 18.8%**

Peer-Reviewed
Conference

2024

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “ChronoGAN: Supervised and Embedded Generative Adversarial Networks for Time Series Generation,” *International Conference on Machine Learning and Applications (ICMLA)*, Sep. 2024. [Online]. Available: <https://doi.org/10.1109/ICMLA61862.2024.00083>, **acceptance rate: 24.3%**

Peer-Reviewed
Conference

2024

M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, “Enhancing Multivariate Time Series-based Solar Flare Prediction with Multifaceted Preprocessing and Contrastive Learning,” *International Conference on Machine Learning and Applications (ICMLA)*, Sep. 2024. [Online]. Available: <https://doi.org/10.1109/ICMLA61862.2024.00056>, **acceptance rate: 24.3%**

Peer-Reviewed
Conference

2024

EskandariNasab, MohammadReza, “Supervised Generative Adversarial Networks for Time Series Generation in Embedding Space” (2024). All Graduate Theses and Dissertations, Fall 2023 to Present. 387.
<https://digitalcommons.usu.edu/etd2023/387>

Thesis

Honors & Awards

IEEE ICDM Conference Travel Award

Nov 2025

- Received a competitive \$500 travel award from IEEE ICDM 2025, funded by the National Science Foundation (NSF), and granted to a limited number of students across the United States.

Contributor to NSF Award #2530946 (CAIG Project)

Jul 2025

- Contributed to the accepted NSF RISE ICER proposal (\$749,989, Utah State University) by proposing UNISON, a novel multimodal generative model with new loss functions and architectures for space weather time series.

SIAM SDM 2025 Student Travel Award

May 2025

- Awarded a highly competitive \$800 travel grant, free conference registration, and SIAM membership by the Society for Industrial and Applied Mathematics (SIAM).

Awarded Thesis-Based M.Sc. in Computer Science

Dec 2024

- Awarded a Master’s degree midway through PhD studies at Utah State University in recognition of outstanding research contributions and academic excellence.

IEEE BigData Conference Travel Award

Dec 2024

- Awarded a competitive \$800 travel grant from IEEE BigData 2024 Conference, funded by the NSF. This recognition was given to only 35 students across the United States for their academic achievements and contributions to the field.

Graduate Student Travel Award, Utah State University

2024–2025

- Received two Graduate Student Travel Awards (\$300 each) from Utah State University to support travel to IEEE BigData 2024 and IEEE ICDM 2025 in Washington, DC, for presenting accepted first-author papers.

NSF Grant Award for SHINE 2024 Workshop Presentation

Aug 2024

- Received an NSF grant award to attend the SHINE 2024 Workshop in Alaska, fully covering a 7-day participation. Delivered the opening talk on the impacts of data preprocessing on solar flare prediction at the Workshop. Presented a poster demonstrating how to achieve a Mean TSS of 0.835 in solar flare classification.

NSF-Funded Research Assistantship Award

Oct 2022

- Awarded an NSF-funded Research Assistantship to develop a machine learning-based web service that enables AI model training for solar flare prediction without coding knowledge. The project was funded by the NSF through the Office of Advanced Cyberinfrastructure (OAC), with a total award amount of \$174,984 (Award #2305781).

Direct PhD Admission at Utah State University

Oct 2022

- Admitted into Utah State University’s direct-entry PhD program through interview and high test scores to study computer science and conduct research on time series analysis and solar flare prediction.

Talks & Posters

IEEE International Conference on Data Mining (ICDM 2025), Washington, D.C.

ORAL PRESENTATION

Nov 2025

- Presented the first-author paper titled *TIMED* in an oral session at IEEE ICDM 2025.

SIAM International Conference on Data Mining (SDM 2025), Alexandria, Virginia

ORAL PRESENTATION

May 2025

- Delivered an oral presentation of the first-author paper titled *AVATAR*, accepted at SIAM SDM 2025.

Utah State University, Utah

PH.D. QUALIFYING EXAMINATION

Mar 2025

- Delivered an oral qualification presentation, presenting two recent research papers and responding to questions from a five-member committee to advance to Ph.D. candidacy.

IEEE International Conference on BigData (BigData 2024), Washington, D.C.

ORAL PRESENTATION

Dec 2024

- Delivered an oral presentation of the first-author paper titled *SeriesGAN*, accepted at IEEE BigData 2024.

IEEE ICMLA 2024, Miami, Florida

VIRTUAL ORAL PRESENTATION

Dec 2024

- Presented two first-author papers accepted as regular papers at IEEE ICMLA 2024 through pre-recorded video presentations.

Utah State University, Utah

MASTER'S THESIS DEFENSE

Nov 2024

- Delivered an oral defense of the master's thesis titled *Supervised Generative Adversarial Networks for Time Series Generation in Embedding Space*.

SHINE 2024 Workshop, Juneau, Alaska

SESSION SPEAKER & POSTER PRESENTATION

Aug 2024

- Delivered a session talk at the SHINE 2024 Workshop on the impacts of data preprocessing on solar flare prediction in the session *Machine Learning-Based Predictions of Solar Flares and SEP Events*.
- Presented a poster showcasing state-of-the-art methods to achieve a Mean TSS of 0.835 in solar flare prediction.

Utah State University, Utah

INVITED GUEST SPEAKER

Jun 2024

- Selected by the Computer Science Department to present data science concepts and solar flare prediction techniques to Indigenous high school seniors, inspiring interest in STEM careers.

Academic & Community Service

IEEE International Conference on Data Mining (IEEE ICDM)

CONFERENCE VOLUNTEER

Nov 2025

- Provided help desk support at IEEE ICDM 2025 in Washington, DC, assisting attendees with room directions, session navigation, and general conference inquiries.

IEEE International Conference on Big Data (IEEE BigData)

CONFERENCE VOLUNTEER

Dec 2024

- Volunteered at IEEE BigData 2024 in Washington, DC, providing technical and logistical support to session chairs and assisting with session coordination to ensure smooth conference operations.

Biomedical Signal Processing and Control (Elsevier)

PEER REVIEWER

Jul 2024 - Now

- Contributed to the field by completing peer reviews of 15 manuscripts for the journal (Impact Factor: 4.9).

Journal of Neural Engineering (IOPscience)

PEER REVIEWER

Sep 2025 - Now

- Completed peer review of three manuscripts for the journal (Impact Factor: 3.8)

Artificial Intelligence (Elsevier)

PEER REVIEWER

Nov 2025

- Conducted peer review of one manuscript for the journal (Impact Factor: 4.6)

Information Fusion (Elsevier)

PEER REVIEWER

Nov 2025

- Completed peer review of one manuscript (Impact Factor: 15.5)

IEEE Journal of Biomedical and Health Informatics (IEEE)

PEER REVIEWER

Dec 2025

- Conducted peer review of one manuscript for the journal (Impact Factor: 6.8)

Astronomy and Computing (Elsevier)

PEER REVIEWER

Nov 2025 – Now

- Served as a peer reviewer for two manuscripts (Impact Factor: 1.6)

Neurocomputing (Elsevier)

PEER REVIEWER

Nov 2025

- Served as a peer reviewer for one manuscript (Impact Factor: 6.5)

Biomedical Physics & Engineering Express (IOPscience)

PEER REVIEWER

Dec 2025

- Completed peer review of one manuscript (Impact Factor: 1.6)

Internet of Things and Cyber-Physical Systems (KeAi Communications Co.)

PEER REVIEWER

Nov 2025

- Conducted peer review of one manuscript for the journal (CiteScore: 37.8)

Information Sciences (Elsevier)

PEER REVIEWER

Sep 2025

- Contributed to the field by completing a peer review for the journal (Impact Factor: 6.8).

Book Chapter: Recent Advances in Deep Learning

PEER REVIEWER

June 2025

- Completed a peer review for a chapter in *Recent Advances in Deep Learning*.

Professional Memberships

Society for Industrial and Applied Mathematics (SIAM)

STUDENT MEMBER

Apr 2025 - Now

- Membership awarded as part of the SIAM SDM 2025 Student Travel Award package.

Institute of Electrical and Electronics Engineers (IEEE)

STUDENT MEMBER

Nov 2024 - Now

Association for Computing Machinery (ACM)

STUDENT MEMBER

Dec 2024 - Dec 2025

Graduate Courses

Spring 2025	Machine Learning for Data Science (score: 4.0/4.0)	Utah State University
Spring 2025	Data-Driven Programming Languages Processing (4.0/4)	Utah State University
Fall 2024	Introduction to Database Systems (score: 4.0/4.0)	Utah State University
Fall 2024	Introduction to Data Analysis (score: 4.0/4.0)	Utah State University
Spring 2024	Applied Deep Learning (score: 4.0/4.0)	Utah State University
Spring 2024	Advanced Data Mining (score: 4.0/4.0)	Utah State University
Fall 2023	Machine Learning with Graphs (score: 4.0/4.0)	Utah State University
Fall 2023	Time Series Data Mining (score: 4.0/4.0)	Utah State University
Summer 2023	Human Factors in Computing (score: 4.0/4.0)	Utah State University

Skills

Languages	Python, C++, SQL
Data Science	TensorFlow, Keras, Scikit-Learn, sktime, Pandas, NumPy, Matplotlib, Seaborn
Neural Networks	GRU, LSTM, RNN, Autoencoders, CNN, ResNet, U-Net
Generative AI	GANs, Diffusion Models, Adversarial Autoencoders, LLMs
Reserach and Coding	LaTeX, BibTeX, Overleaf, Jupyter Notebook, Cuda, Visual Studio Code
Databases	PostgreSQL, MongoDB, Redis
Web Development	Django, REST APIs, React.js, Celery, HTML, CSS, Nginx, Waitress, Git

Datasets and Notebooks

2024	M. EskandariNasab , S. M. Hamdi, and S. Filali Boubrahimi, "Cleaned SWANSF Dataset (v1.0.0)," <i>Zenodo</i> , Jun. 11, 2024. doi: https://doi.org/10.5281/zenodo.11566472 .	Dataset
2024	M. EskandariNasab , S. M. Hamdi, and S. Filali Boubrahimi, "SWAN-SF Data Preprocessing and Sampling Notebooks (v1.0.0)," <i>Zenodo</i> , Jun. 11, 2024. doi: https://doi.org/10.5281/zenodo.11564789 .	Computational Notebook

Independent Projects

Lung Segmentation in Chest X-Ray Dataset using U-Net PYTHON, KERAS, U-NET MODEL, NUMPY, PANDAS, AND MATPLOTLIB https://github.com/samresume/Lung-Segmentation-using-U-Net	2022
Detecting COVID-19 from X-Ray Images using CNNs PYTHON, KERAS, AND NUMPY https://github.com/samresume/Detecting-COVID-19-using-CNNs	2021
Brain Tumor Detection from MRI Images using CNNs PYTHON, KERAS, AND NUMPY https://github.com/samresume/Brain-Tumor-Detection-using-CNNs	2021

Detecting Anomalies in Patient Heartbeats using LSTM Autoencoders PYTHON, PYTORCH, PANDAS, MATPLOTLIB, AND SEABORN https://github.com/samresume/Detecting-Anomalies-using-LSTM-Autoencoder	2022
Forecasting Weather using Australian Weather Data and Neural Prophet PYTHON AND PANDAS https://github.com/samresume/Forecasting-Weather-using-Neural-Prophet	2021
COVID-19 Tweets Sentiment Analysis using BERT Transformer PYTHON, PYTORCH, BERT MODEL, PANDAS, NUMPY, MATPLOTLIB, AND SEABORN https://github.com/samresume/COVID-19-Tweets-Analysis-using-BERT	2022
Automated Stocks and Crypto News Analysis using NLP PYTHON, FINE-TUNED PEGASUS TRANSFORMER MODEL, TRANSFORMERS PIPELINES, AND PANDAS https://github.com/samresume/Automated-Stock-and-Crypto-News-Analysis	2022
Exploratory Data Analysis on Chronic Kidney Disease Dataset PYTHON, PANDAS, MATPLOTLIB, AND SEABORN https://github.com/samresume/Exploratory-Data-Analysis-on-Kidney-Disease-Dataset	2022

Test Scores

TOEFL iBT	Jul 2022
99 (R: 29, L: 28, S: 21, W: 21)	

Languages

English	Persian
Full professional proficiency	Native or bilingual proficiency

References

Shah Muhammad Hamdi ASSISTANT PROFESSOR, UTAH STATE UNIVERSITY - s.hamdi@usu.edu +1 (435) 797 1573 - SER 330, Science Engineering Research Building, Logan, UT 84321	<i>Ph.D. Supervisor</i>
Soukaina Filali Boubrahimi ASSOCIATE PROFESSOR, UTAH STATE UNIVERSITY - soukaina.boubrahimi@usu.edu +1 (435) 797 1020 - SER 332, Science Engineering Research Building, Logan, UT 84321	<i>Faculty Mentor</i>
Hamid Karimi ASSISTANT PROFESSOR, UTAH STATE UNIVERSITY - hamid.karimi@usu.edu +1 435 797 1233 - SER 326, Science Engineering Research Building, Logan, UT 84321	<i>Reference Professor</i>